

Cloud Monitoring



Cloud monitoring helps you collect, visualize and alert on metrics, logs and events from cloud resources to ensure performance, availability, security and cost optimization.

1 AWS CloudWatch Exporter



AWS
CloudWatch

- Exports AWS CloudWatch metrics to Prometheus.
- Supports various services for easy discovery.
- Works with EC2, EKS, RDS, ALB, S3 and more.

Key Metrics

- CPUUtilization
- DiskRead/Write
- NetworkIn/Out
- StatusCheckFailed
- RequestCount
- TargetResponseTime
- Custom Metrics

Example scrape config

```
- job_name: 'aws-cloudwatch'
  cloudwatch_sd_configs:
    - region: 'ap-south-1'
      metrics:
        - namespace: AWS/EC2
          namefilters_name: CPUUtilization
          ec2_dimensions: [InstanceId]
          stats: [Average: [Average]]
          period: 60
```

How it Works



CloudWatch has
metrics from
AWS services



CloudWatch
Exporter pulls
metrics via API



Metrics are
exposed in
Prometheus format



Use Grafana
to visualize and
create alerts

Use Cases

- ✓ Monitor AWS infrastructure and services
- ✓ Create dashboards and alerts in Grafana
- ✓ Track performance, availability and errors

Cloud Monitoring



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2 Azure Monitor Exporter



Azure
Monitor

- Exports Azure Monitor metrics to Prometheus.
- Supports auto discovery using Azure tags and resource groups.
- Works with VMs, AKS, App Services, Storage, SQL, etc.

Key Metrics

- CPU Percentage
- Memory Usage
- Disk Read/Write
- Network In/Out
- Requests
- Failed Requests
- Availability

Example scrape config

```
- job_name: 'azure-monitor'
  azure_sd_configs:
    - subscription_id: 'xxxxxxxxx-xxxx'
      tenant_id: 'xxxxxxxxx-xxxx'
      client_id: 'xxxxxxxxx-xxxx'
      client_secret: 'xxxxxxxxx-xxxx'
```

How it Works



Azure
Resources



Azure Monitor
collects metrics



Azure Monitor
Exporter pulls
metrics via API



Metrics are
exported in
Prometheus format



Use Grafana
to visualize and
create alerts

Use Cases

- ✓ Monitor Azure VMs, AKS, App Services, Storage, SQL and more
- ✓ Create dashboards and alerts in Grafana
- ✓ Track performance, availability and errors
- ✓ Centralize monitoring for all Azure resources

Tips



- Use Managed Identity instead of client secret for better security.
- Filter metrics using tags and resource groups.
- Combine with Alertmanager for advanced alert routing.



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3 GCP Monitoring



Google Cloud Monitoring

- GCP metrics are exported using google-cloud-monitoring exporter.
- Supports projects, zones, and instances.
- Integrates with Cloud Monitoring & Prometheus.

Key Metrics

- CPU Utilization
- Disk Throughput
- Network Throughput
- Instance Count
- Load Balancer Metrics
- Custom Metrics
- Uptime Check

Example scrape config

```
- job_name: 'gcp-monitoring'
  google_cloud_monitoring_scrape_configs:
    - project_id: 'your-project-id'
      metrics_prefix: 'custom.googleapis.com'
      filter: 'resource.type=gce_instance'
```

How it Works



Tools to Use



Prometheus



gcp-monitoring
exporter



Cloud Advisor



CloudWatch
(For logs)



Grafana

Use Cases

- ✓ Monitor GCP VMs, GKE, App Engine, Cloud SQL, etc.
- ✓ Create dashboards and alerts in Grafana
- ✓ Track performance, availability and errors
- ✓ Monitor load balancers, uptime checks and custom metrics

Tips

- Use service accounts with least privilege access.
- Filter metrics to reduce cardinality and cost.
- Combine with Alertmanager for smart alert routing.

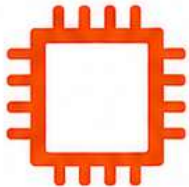


Cloud Monitoring



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4 EC2 Monitoring



EC2 Monitoring

- Monitor EC2 instances using CloudWatch metrics or Node Exporter.
- Key metrics include CPU, Memory, Disk, Network and Status Checks.
- Use detailed monitoring (1-min interval) for better visibility.

Important Metrics

- CPUUtilization
- DiskReadBytes
- DiskWriteBytes
- NetworkIn
- NetworkOut
- StatusCheckFailed
- CPUCreditBalance

Example scrape config

```
- job_name: 'ec2-monitoring'
  metrics_path: /metrics
  static_configs:
    - targets: ['10.0.1.10:9100', '10.0.1.11:9100'] # Node Exporter on EC2
      labels:
        env: 'production'
        role: 'web-server'
```

Best Practices

- ✓ Enable detailed monitoring for production instances.
- ✓ Set CloudWatch alarms for critical metrics.
- ✓ Use dashboards to visualize instance health.

How it Works



EC2 Instance
(Workload)



CloudWatch /
Node Exporter
collects metrics



Metrics are
exported in
Prometheus format

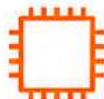


Use Grafana to
visualize and
create alerts

Tools to Use



AWS
CloudWatch



EC2
Instances



kube-state-
metrics



Prometheus



Grafana



CloudWatch
Alarms

Use Cases

- ✓ Monitor EC2 instance performance and health
- ✓ Track CPU, Memory, Disk and Network usage
- ✓ Set alerts for high utilization or failures
- ✓ Monitor status checks and instance availability

Tips

- Use Auto Scaling group metrics for fleet monitoring.
- Combine CloudWatch metrics with Node Exporter for granular visibility.
- Create dashboards for business critical instances.
- Enable detailed monitoring (1-minute) for accuracy.



Cloud Monitoring



Cloud monitoring helps you collect, visualize and alert on metrics, logs and events from cloud resources to ensure performance, availability, security and cost optimization.

5 EKS Monitoring



EKS
Monitoring

- Monitor EKS cluster, nodes, pods and services.
- Use CloudWatch, kube-state-metrics, Advisor and Prometheus.
- Track control plane metrics for node health.

Key Metrics

- Node CPU / Memory
- Pod Status
- Container CPU / Memory
- Kube API Latency
- Cluster Autoscaler
- Scheduler Performance
- ETCD Health

Tools to Use



Prometheus



kube-state-metrics



Advisor



CloudWatch



Grafana

Best Practices

- ✓ Monitor cluster and node health continuously.
- ✓ Set alerts for pod failures, high CPU/Memory and API latency.
- ✓ Use dashboards to visualize cluster performance.

How it Works



EKS Cluster
(Nodes, Pods,
Services)



kube-state-metrics
collects k8s
object metrics



Prometheus scrapes
metrics from
exporters



CloudWatch collects
infrastructure &
control plane metrics



Use Grafana to
visualize and
create alerts

Use Cases

- ✓ Monitor EKS cluster, nodes, pods and services
- ✓ Track control plane and system components
- ✓ Visualize performance and resource usage
- ✓ Set up alerts for failures and high utilization
- ✓ Ensure availability and health of workloads

Tips

- Deploy Prometheus and kube-state-metrics using Helm charts.
- Use Prometheus Adapter for custom metrics in HPA.
- Enable Cluster Autoscaler and monitor scaling events.
- Keep dashboards simple and focused on important metrics.





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6 Multi-Cloud Monitoring

- Monitor resources across AWS, Azure and GCP in a single platform.
- Centralize metrics using Prometheus and visualize in Grafana.
- Ensure consistent alerting, dashboards and naming conventions.

Tools to Use



Prometheus

Collects and stores metrics from all clouds.



Grafana

Creates dashboards and visualizations.



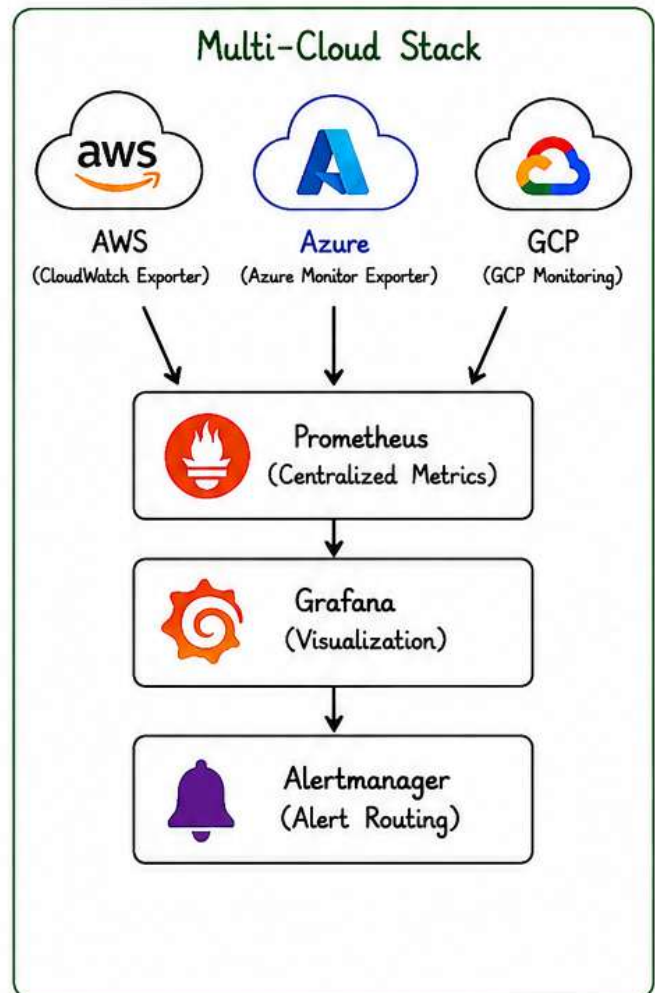
Alertmanager

Handles alerts and notifications.



Exporters

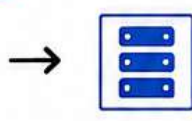
CloudWatch, Azure Monitor and GCP exporters.



How it Works



Resources in AWS, Azure and GCP



Exporters collect metrics from each cloud provider



Metrics are sent to Prometheus for storage



Grafana visualizes metrics in unified dashboards



Alertmanager sends alerts via Email, Slack, Telegram etc.

Use Cases

- ✓ Monitor workloads across multiple cloud providers
- ✓ Create unified dashboards for all environments
- ✓ Track performance, availability and cost across clouds
- ✓ Centralize alerting and notifications
- ✓ Ensure consistency in monitoring and observability

Best Practices

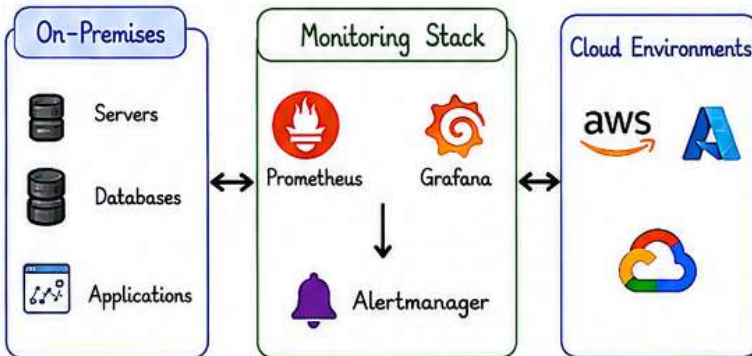
- Use Prometheus as the central metrics store.
- Follow consistent naming and labeling for metrics.
- Use recording rules to reduce query load.
- Create role-based dashboards for different teams.
- Test alerts and notification channels regularly.





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7 Hybrid Cloud Monitoring



Best Practices

- ★ Use a unified monitoring platform for all environments.
- ★ Standardize metric names and labels.
- ★ Store monitoring data on-prem and cloud.
- ★ Follow security, best practices and application SLAs.
- ★ Ensure high availability of monitoring components.
- ★ Test alerts and dashboards regularly.
- ★ Document monitoring architecture and runbooks.

Monitoring Components

	Prometheus	Collects metrics from all systems (on-prem & cloud).
	Grafana	Visualizes metrics using dashboards.
	Alertmanager	Handles alerts and notifications.
	Exporters	Collects metrics from systems (Node Exporter, CloudWatch Exporter, etc.).

Tools to Use

	Prometheus	Time series database for metrics storage.
	Grafana	Dashboarding and visualization tool.
	Alertmanager	Alert routing and notification manager.
	Exporters	Collectors for metrics from different sources.
	Blackbox Exporter	For uptime and endpoint monitoring.
	ELK / Loki	For log collection and analysis.

How it Works



Use Cases

- ✓ Monitor on-prem servers and cloud resources together
- ✓ Track application performance across environments
- ✓ Ensure availability and uptime of critical services
- ✓ Centralize logs, metrics and alerts
- ✓ Cost and resource optimization across hybrid setup
- ✓ Improve security and compliance monitoring

Tips

- Use federation in Prometheus for multi-environment setup.
- Manage secrets and access using roles and policies.
- Use consistent naming, tagging and labeling.
- Keep dashboards clean, simple and meaningful.
- Backup monitoring configuration and dashboards.
- Continuously review and optimize alert rules.

